

HTML Audio & Video Controls

- ▶ Understanding HTML5 Media Elements

Introduction to Media Elements

- ▶ - HTML5 introduced <audio> and <video>
- ▶ - Allows embedding media directly in browser
- ▶ - Supports built-in controls for playback
- ▶ - No extra plugins required

Audio Element

- ▶ - <audio> tag embeds audio files
- ▶ - Common attributes:
 - ▶ - controls
 - ▶ - autoplay
 - ▶ - loop
 - ▶ - muted
 - ▶ - preload

Example Code:

```
<audio controls>  
  <source src="sample-audio.mp3" type="audio/mpeg">  
  Your browser does not support the audio element.  
</audio>
```

Video Element

- ▶ - <video> tag embeds video files
- ▶ - Common attributes:
- ▶ - controls
- ▶ - autoplay
- ▶ - loop
- ▶ - muted
- ▶ - preload
- ▶ - poster

▶ Example Code:

```
<video width="480" height="270" controls>
```

```
<source src="sample-video.mp4" type="video/mp4">
```

Your browser does not support the video element.

```
</video>
```

Audio Attributes

- ▶ - controls: display play/pause/volume UI
 - ▶ - autoplay: start automatically
 - ▶ - loop: repeat playback
 - ▶ - muted: start muted
 - ▶ - preload: auto, metadata, none
- ▶ Example:
- ```
<audio controls autoplay loop muted preload="auto">
<source src="sample-audio.mp3" type="audio/mpeg">
</audio>
```

# Video Attributes

- ▶ - controls: show playback UI
- ▶ - autoplay: start video automatically
- ▶ - loop: repeat video
- ▶ - muted: start muted
- ▶ - preload: auto, metadata, none
- ▶ - poster: image shown before video plays

▶ Example:

```
<video width="480" height="270" controls autoplay loop muted
poster="poster.jpg">
```

```
<source src="sample-video.mp4" type="video/mp4">
```

```
</video>
```

# Preload Attribute

- ▶ - auto: load full media before playback
- ▶ - metadata: load only info (duration, dimensions)
- ▶ - none: load nothing until user plays
- ▶ - Helps improve performance and bandwidth usage



# Preload: metadata

```
<video controls preload="metadata">
```

```
<source src="sample-video.mp4" type="video/mp4">
```

```
</video>
```

- ▶ **browser Internally reads:**
  - ▶ “This video is 2 min 30 sec”
  - ▶ “Width = 1280, Height = 720”
- ▶ Does **NOT download full video**
- ▶ Shows duration on control bar

# Preload: Auto

- ▶ Preload="auto" tells the browser:  
“**Start loading the media file as much as you think is needed, as soon as the page loads.**”
- ▶ It is a **hint**, not a strict command — the browser may still optimize.
- ▶ When the page loads:
- ▶ Browser sees <audio> or <video>
- ▶ Sees preload="auto"
- ▶ Browser:
  - ▶ downloads **metadata first**
  - ▶ then **starts downloading media data**
- ▶ Result:
  - ▶ Media is **ready to play instantly**
  - ▶ Little or no waiting when user clicks

# Example

```
<video width="400" controls preload="auto">
<source src="movie.mp4" type="video/mp4">
</video>
```

# Summary

- ▶ - <audio> and <video> embed media in HTML5
- ▶ - controls attribute adds built-in UI
- ▶ - Other attributes modify playback behavior
- ▶ - Use <source> for multiple formats
- ▶ - preload optimizes loading for better UX

# Class Task 7

- ▶ - Add an audio element with controls, autoplay, and loop
- ▶ - Add a video element with poster, muted, and preload="metadata"
- ▶ - Test in different browsers